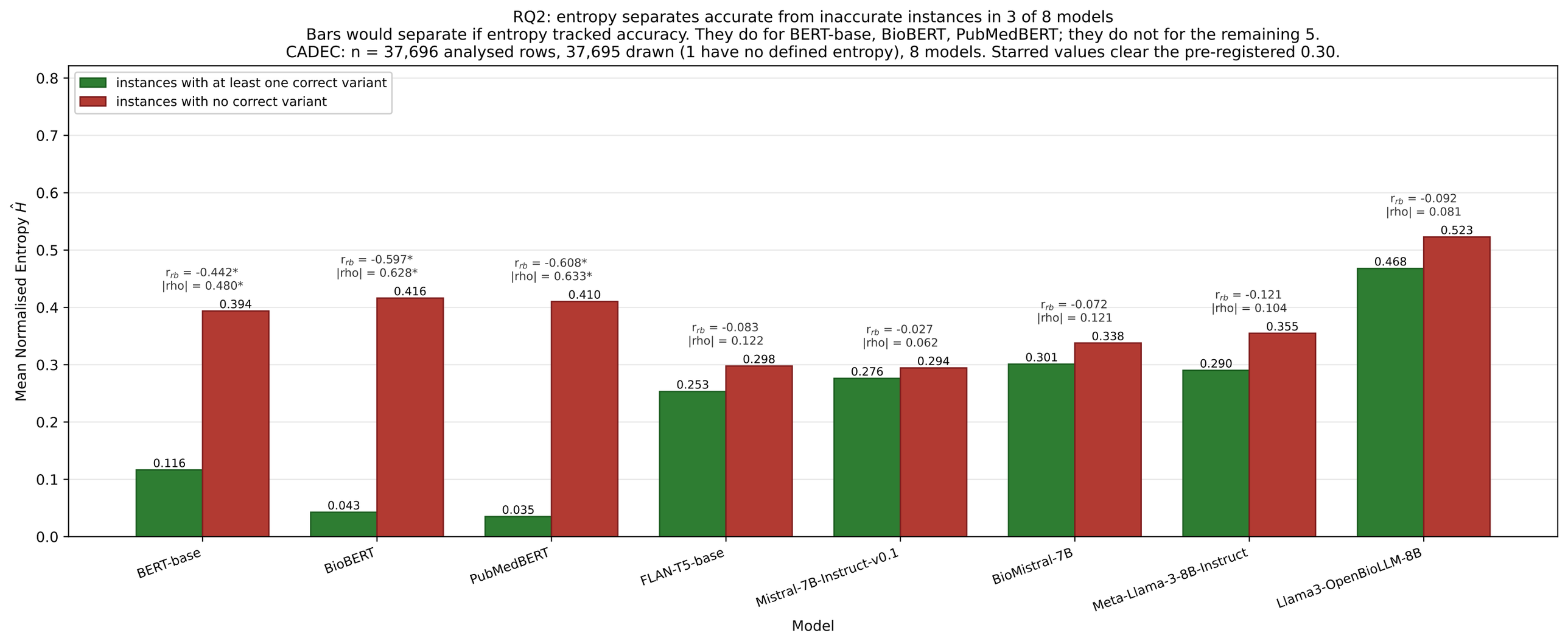
RQ2: entropy separates accurate from inaccurate instances in 3 of 8 models
Bars would separate if entropy tracked accuracy. They do for BERT-base, BioBERT, PubMedBERT; they do not for the remaining 5.
CADEC: n = 37,696 analysed rows, 37,695 drawn (1 have no defined entropy), 8 models. Starred values clear the pre-registered 0.30.
instances with at least one correct variant
instances with no correct variant
Mean Normalised Entropy $\hat{H}$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
r_{rb} = -0.442*
|rho| = 0.480*
0.116
0.394
r_{rb} = -0.597*
|rho| = 0.628*
0.043
0.416
r_{rb} = -0.608*
|rho| = 0.633*
0.035
0.410
r_{rb} = -0.083
|rho| = 0.122
0.253
0.298
r_{rb} = -0.027
|rho| = 0.062
0.276
0.294
r_{rb} = -0.072
|rho| = 0.121
0.301
0.338
r_{rb} = -0.121
|rho| = 0.104
0.290
0.355
r_{rb} = -0.092
|rho| = 0.081
0.468
0.523
BERT-base
BioBERT
PubMedBERT
FLAN-T5-base
Mistral-7B-Instruct-v0.1
BioMistral-7B
Meta-Llama-3-8B-Instruct
Llama3-OpenBioLLM-8B
Model